Raphaël URFIN

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About me

I am a second year PhD student at École Normale Supérieure - PSL in Paris supervised by Giulio Biroli (ENS) and Marc Mézard (Bocconi University) working on the theory of diffusion models. My background is in theoretical physics.

Education

- **PhD in Physics and Machine Learning** — **ENS-PSL, Paris**
  Subject: Generative Diffusion and Statistical Physics — *2025 – 2028 (exp.)*
  LPENS & Centre de Sciences des Données
  Supervisors: Giulio Biroli (ENS) and Marc Mézard (Bocconi University)
- **ENS Diploma** — **ENS-PSL, Paris**
  Interdisciplinary Diploma validating my studies at ENS. — *2021 – 2025*
- **M2 ICFP Theoretical Physics Track** — **ENS-PSL, Paris**
  First Class Honors, 17.38/20 — *2023 – 2024*
  Relevant classes: Advanced Statistical Physics, Disordered Systems, Machine Learning.
- **M1 ICFP** — **ENS-PSL, Paris**
  First Class Honors, 17.8/20 — *2022 – 2023*
  Relevant classes: Phase Transitions, Introduction to Quantum Field Theory, General Relativity.
- **Bachelor in Physics and Mathematics** — **ENS-PSL, Paris**
  First Class Honors, 17.38/20 (Physics) and 16.36/20 (Mathematics) — *2021 – 2022*

Publications

* denotes equal contribution.

Preprints

- **Raphaël Urfin**, Tony Bonnaire, Giulio Biroli, Marc Mézard.
  First Learn, Then Memorize: The Spectral Bias of Diffusion Models.
  arXiv:2609.35377 (2026). https://arxiv.org/abs/2609.35377
- Jérémie Klinger, **Raphaël Urfin**, Giulio Biroli, Marylou Gabrié.
  Weighting Schedules Govern What and When Score-Based Generative Models Learn from Multimodal Data.
  arXiv:2609.35322 (2026). https://arxiv.org/abs/2609.35322
- **Raphaël Urfin***, Tony Bonnaire*, Giulio Biroli, Marc Mézard.
  Double Descent and Malign Overfitting in Diffusion Models.
  arXiv:2609.26392 (2026). https://arxiv.org/abs/2609.26392

Conference Proceedings

- Tony Bonnaire*, **Raphaël Urfin***, Giulio Biroli, Marc Mézard.
  Why Diffusion Models Don't Memorize: The Role of Implicit Dynamical Regularization in Training.
  NeurIPS 2025, **Best Paper Award (4 papers out of 20,000+ submissions)**
  Advances in Neural Information Processing Systems (NeurIPS) 38, 141266-141286
  Selected for the JSTAT special issue *Machine Learning 2026*: *J. Stat. Mech.* (2026) 084007.

Conferences, Workshops and Seminars

Invited Talks

- **Generative Memory Lab, Donders Institute** — **Radboud University, Netherlands (Online)**
  Invited Seminar — *September 2026*
- **LemanTH 2026** — **Lausanne, Switzerland**
  Workshop — *April 2026*
- **Vitelli's Group Meeting** — **Online**
  Group meeting (non-reciprocal p-spins) — *March 2026*
- **Scattering Club, Data Science for Astrophysics group @ LPENS** — **Online**
  Invited Seminar (1 hour) — *February 2026*
- **Voxel51 Best of NeurIPS 2025** — **Online**
  Invited Seminar (30 min) — *January 2026*
- **Centre de Sciences des Données Internal Seminar** — **Paris, France**
  Invited Seminar (30 min) — *December 2025*
- **Machine Learning & Signal Processing @ ENS Lyon** — **Lyon, France**
  Invited Seminar (1 hour) — *November 2025*

Contributed Talks

- **Journées de Physique Statistique 2026** — **Paris, France**
  Conference (4-min flash talk) — *January 2026*
- **EurIPS 2025** — **Copenhagen, Denmark**
  Conference (poster + 13-min oral) — *December 2025*
- **NeurIPS in Paris 2025** — **Paris, France**
  Workshop (poster + 10-min oral) — *November 2025*
- **StatPhys 29** — **Florence, Italy**
  Conference (15-min talk) — *July 2025*
- **Youth in High Dimensions** — **Trieste, Italy**
  Workshop (15-min talk) — *July 2025*
- **Journées de Physique Statistique 2025** — **Paris, France**
  Conference (4-min flash talk) — *January 2025*

Summer Schools

- **Beg Rohu Summer School of Statistical Physics** — **Beg Rohu, France**
  Summer School — *June 2025 & June 2026*
  - **2026 Edition (Optimization and Control of Nonequilibrium Systems):** Lectures by international researchers (e.g. G. Peyré, G. Rotskoff ...). Presented a poster on Memorization/Generalization in diffusion models.
  - **2025 Edition (Learning with Machines, Physics and Mind):** Lectures on Machine Learning and Statistical Physics by international researchers (e.g., Yann LeCun, Julia Kempe, Stéphane Mallat, Marc Mézard ...). Presented a poster on Memorization/Generalization in diffusion models.
- **Complex and Glassy Systems** — **Cargèse, France**
  Summer School — *July 2024*
  - Lectures on Statistical Physics and interdisciplinary applications by international researchers (e.g. Marc Mézard, Eric Vanden-Eijnden, Valentina Ros, Guy Bunin).

Awards and Grants

- 2025: Recipient of one of the four best paper awards at NeurIPS 2025. **(4 papers out of 20,000+ submissions)**
- 2025: CDSN, Competitive PhD scholarship for ENS students

Academic Activities

Teaching

- TA Phase Transitions (M1 level, Master ICFP, ENS-PSL), 16 hours, taught in English with Giulio Biroli. Fall 2026.
- TA Mathematics Tutoring for first-year students in Physics at ENS, 36 hours, taught in French with Amir-Kian Kashani-Poor. Fall 2025 and Fall 2026.
- TA Machine Learning (M2, Master ICFP, ENS-PSL), 11 hours, taught in English with Marc Lelarge and Leonardo Defilippis. Spring 2026.
- TA Stochastic Processes for Physics (M1 level, Master ICFP, ENS-PSL), 16 hours, taught in English with Marylou Gabrié. Fall 2025.
- "Khôlles" (Preparation for the oral exams of French "Grandes Écoles") in Mathematics and Physics for Classes Préparatoires PC, Lycée Stanislas (2022–2024).

Student Supervision

- **Sacha Gaucher** — **ESPCI Paris**
  Master's Research Internship — *April – July 2026*
  - Co-supervised with Giulio Biroli.
  - Subject: The RePaint method (inpainting with diffusion models).

Reviewing

- Machine Learning Workshops: EurIPS 2025 Workshop Principles of Generative Modeling (PriGM), ICLR 2026 Workshop DeLTa, ICLR 2026 Workshop Sci4DL.
- Machine Learning Journals: TMLR

Skills

- Languages: French (Native), English (Fluent), Italian (Fluent).
- Software skills: Python (Pytorch), Matlab, Latex.

Relevant Experience

- **Statistical Physics and Diffusion Models** — **Bocconi University, Milan, Italy**
  Research Internship — *February – May 2025*
  - Supervisor: Marc Mézard, Department of Computing Sciences.
  - Subject: The Memorization/Generalization transition in diffusion models.
  - Resulted in a publication accepted at NeurIPS 2025
- **Statistical Physics and Diffusion Models** — **ENS-PSL, Paris**
  Research Internship — *April 2024 – January 2025*
  - Supervisor: Giulio Biroli, Centre de Sciences des Données.
  - Subject: The effect of implicit regularization in diffusion models.
- **Emergent behaviors in large ecosystems** — **University of Cambridge, UK**
  Research Internship — *January – July 2023*
  - Supervisor: Camille Scalliet, Soft Matter Group, Department of Applied Mathematics and Theoretical Physics.
  - Using tools from disordered systems (Cavity Method, Random Matrix Theory...) to understand emergent collective behaviors in ecological systems.